Ramm's finite-rank problems for transverse affine hyperplanes

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Status. Likely valid partial result; the general curved-variety questions remain open.

Source and exact question

A. G. Ramm, "Three open problems in analysis," arXiv:math/0603631 (2006), Problems 3.1 and 3.2. Let

$$F(z)=\int_D f(x)e^{iz\cdot x}\,dx, \qquad z\in\mathbb{C}^n,$$

where $D\subset\mathbb{R}^n$ is bounded and $0\neq f\in L^2(D)$. For two transverse algebraic hypersurfaces $\mathcal{L}_1,\mathcal{L}_2$, Problem 3.1 asks whether the integral operator with kernel $F(u+v)$ has infinite rank. Problem 3.2 asks whether, for every M, one can choose $p_1,\dots,p_M\in\mathcal{L}_2$ so that the functions $F(z+p_m)|_{\mathcal{L}_1}$ are linearly independent.

counterexample.

In the one-dimensional case the answer to the question (**) is yes (see [?]).

3.A problem in operator theory.

The question in two different forms is stated below as Problem 3.1 and Problem 3.2. These problems are closely related.

3.1. Let D be a bounded domain in R^3, D can be a box or a ball, $f(x)\in L^2(D)$ be a function $f\not\equiv 0$, $F(z):=\int_D f(x)\exp(iz\cdot x)dx$, where $z\in C^3$ is an entire function of exponential type. Let $L_j(z)$, $j=1,2$, be polynomials, $\deg L_j(z)\geq 1$, $\mathcal{L}_j:=\{z: z\in C^3, L_j(z)=0\}$ be algebraic varieties.

Define Hilbert spaces $H_j:=L^2(\mathcal{L}_j,dm_j)$, where $dm_j(z)$ are smooth rapidly decaying strictly positive measures on $\mathcal{L}_j$, such that any exponential $exp(iz\cdot x)$ with any $x\in R^3$ belongs to H_j. Define linear operator T from H_1 into H_2 by the formula: $Th:=\int_{\mathcal{L}_1} dm_1 h(u_1)F(u_1+u_2):=g(u_2)$, where $u_j\in\mathcal{L}_j$, $h\in H_1$, $g\in H_2$ for any $h\in H_1$ (such are measures dm_j by the assumption). Assume that $\mathcal{L}_1$ and $\mathcal{L}_2$ are transversal, which by definition means that there exist two points, one in $\mathcal{L}_1$ and one in $\mathcal{L}_2$, such that the union of the bases of the tangent spaces to $\mathcal{L}_1$ and to $\mathcal{L}_2$ at these points, form a basis in C^3. The same setting is of interest in dimension $n>3$ as well.

Problem 3.1: *Is it true that T is not a finite-rank operator?*

That is, is it true that dimension of the range of T is infinite?

Remark: The assumption that $f(x)$ is in $L^2(D)$ is important: if, for example, $f(x)$ is a delta function, then the answer to the question of problem 3.1 is no: the dimension

2

Partial theorem

An affine complex hyperplane is a translate $a + E$ of a codimension-one complex linear subspace $E \subset \mathbb{C}^n$. Two such hyperplanes are called transverse here when their direction spaces satisfy $E_1 + E_2 = \mathbb{C}^n$.

Theorem 1. *Let $n \geq 3$, let $H_j = a_j + E_j \subset \mathbb{C}^n$ be transverse affine complex hyperplanes, and let the smooth positive rapidly decaying measures in Ramm's setup be fixed. If $0 \neq f \in L^2(D)$ for a bounded $D \subset \mathbb{R}^n$, then the operator with analytic kernel*

$$K(u, v) = F(u + v), \qquad (u, v) \in H_1 \times H_2,$$

has infinite rank. Moreover, for every $M \geq 1$ there are $p_1, \dots, p_M \in H_2$ such that $F(\,\cdot + p_1), \dots, F(\,\cdot + p_M)$ are linearly independent on H_1.

Proof intuition

Transverse hyperplanes in dimension at least three share a nonzero tangent direction w. If the kernel had finite rank, its sections would form a finite-dimensional space invariant under translation in that shared direction. The generator of a finite-dimensional translation action obeys its characteristic polynomial. Consequently F obeys a nontrivial constant-coefficient differential equation $P(\partial_w)F = 0$. Fourier inversion turns this into $P(iw \cdot x)f(x) = 0$. A nonzero polynomial in a nonzero linear form vanishes only on a Lebesgue-null set, which is incompatible with a nonzero L^2 density.

Proof

Write $W = E_1 \cap E_2$. Since E_1 and E_2 both have complex dimension $n - 1$ and $E_1 + E_2 = \mathbb{C}^n$,

$$\dim_{\mathbb{C}} W = n - 2 \geq 1.$$

Fix $0 \neq w \in W$.

Suppose first that the kernel operator T has finite rank. For $u \in H_1$, put

$$k_u(v) = F(u + v), \qquad v \in H_2.$$

Every k_u belongs to the finite-dimensional range of T. Indeed, a smooth approximate identity on H_1 concentrated at u, inserted as the input of T, converges to k_u in $L^2(H_2, dm_2)$. The required dominated convergence follows from compactness of D and from the assumed finite H_2-norms of all exponentials; their squared norms are moment-generating functions of the rapidly decaying measure and are locally bounded in the spatial parameter. A finite-dimensional range is closed. Thus

$$V := \operatorname{span}\{k_u : u \in H_1\}$$

is finite-dimensional. (If $V = \{0\}$, then F vanishes on $H_1 + H_2 = \mathbb{C}^n$, already contradicting $f \neq 0$.)

For $t \in \mathbb{C}$, translation in the direction w preserves V, because

$$k_u(v + tw) = F(u + v + tw) = k_{u+tw}(v), \qquad u + tw \in H_1, quad v + tw \in H_2.$$

Equality in $L^2(H_2, dm_2)$ is equality everywhere for these analytic functions, so the translation action and its infinitesimal generator $A = \partial_w|_V$ are well-defined. Let P be the characteristic polynomial of A. Cayley–Hamilton gives

$$P(\partial_w)F(u + v) = P(\partial_w)k_u(v) = 0 \quad (u \in H_1, v \in H_2).$$

Transversality gives $H_1 + H_2 = \mathbb{C}^n$, hence

$$P(\partial_w)F(z) = 0 \qquad (z \in \mathbb{C}^n). \tag{1}$$

Because D is bounded, $f \in L^1(D)$ and differentiation under the integral is legitimate. Equation (1) is the Fourier–Laplace transform of

$$x \longmapsto P(iw \cdot x)f(x).$$

Restricting (1) to $z \in \mathbb{R}^n$ and using uniqueness of the Fourier transform on L^1 yields

$$P(iw \cdot x)f(x) = 0 \quad \text{for a.e. } x. \tag{2}$$

The function $x \mapsto P(iw \cdot x)$ is a nonzero polynomial on $\mathbb{R}^n$. It cannot vanish identically: the image of the nonzero real-linear map $x \mapsto iw \cdot x$ is infinite, whereas a nonzero one-variable polynomial has finitely many zeros. Its zero set therefore has Lebesgue measure zero. Equation (2) forces $f = 0$ almost everywhere, the desired contradiction. Thus T has infinite rank.

Interchanging H_1 and H_2 in the same argument shows that

$$\operatorname{span}\{F(\,\cdot\, + p)|_{H_1} : p \in H_2\}$$

is infinite-dimensional. From any infinite-dimensional family one may select M linearly independent members inductively. This is precisely the conclusion requested in Problem 3.2. $\square$

Verification and scope

The proof uses only: (i) analytic kernel sections lie in the closed finite rank range by approximate identities; (ii) transverse affine hyperplanes have a nonzero common direction when $n \geq 3$; (iii) Cayley–Hamilton; and (iv) uniqueness of the Fourier transform. The restriction $n \geq 3$ is essential to this argument because two transverse affine lines in $\mathbb{C}^2$ have no common direction.

The result does not extend formally to curved algebraic hypersurfaces: their tangent-space intersection at a point does not supply a global translation direction. Problems 1 and 2 in the source are not addressed by the theorem.

Novelty check

A bounded search on 13 August 2026 covered the exact phrases "$F(z + p_m)$ linearly independent," "finite-rank operator algebraic varieties $F(u_1 + u_2)$," the source title and Problems 3.1/3.2, and variants with "transverse affine hyperplanes." It found the source and generic finite-rank/Fourier references, but no later solution or this affine subcase. The appropriate claim is therefore "apparently new within the bounded search," not an exhaustive novelty assertion.

Human-review recommendation

Check chiefly the approximate-identity passage placing every analytic kernel section in $\operatorname{Ran} T$, and the induced translation action on V. The remaining algebraic and Fourier steps are direct.

References

1. A. G. Ramm, "Three open problems in analysis," arXiv:math/0603631 (2006).
2. L. Schwartz, standard uniqueness and Fourier–Laplace theory for compactly supported distributions; only ordinary L^1 Fourier uniqueness is needed in the proof above.